CHA-1.A.1 Calculus uses limits to understand and model dynamic change.

CHA-1.A.2 Because an average rate of change divides the change in one variable by the change in another, the average rate of change is undefined at a point where the change in the independent variable would be zero.

CHA-1.A.3 The limit concept allows us to define instantaneous rate of change in terms of average rates of change.

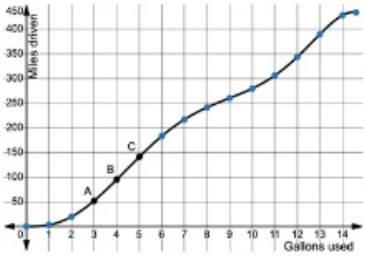
1-1 Change at an Instant

Get Ready...

One factor that people consider when buying a car is fuel efficiency. Some cars guzzle gas while others are considered "eco" and use very little gas. This data can be tracked in the car's display, but how are these numbers calculated and what do they really mean?

1. The graph shows the number of miles driven and gallons of gas used on one Trip.
   a. Estimate the car's average fuel economy on this Trip. Show your work.

$$\frac{430 \text{ miles}}{14.5 \text{ gal}} \approx 29.655 \text{ mi/gal}$$



   b. Estimate the car's average fuel economy between Point A and Point C. Show your work.

$$\frac{145-52 \text{ mi}}{5-3 \text{ gal}} \approx 46.5 \text{ mi/gal}$$

2. Suppose you wanted to estimate the car's instant fuel economy at Point B.
   a. How is this calculation different than finding an average fuel economy?

There is not an interval over which we can calculate the distance driven and the gallons consumed.

   b. Will the answer be closer to the value you found in 1a or 1b? Explain.

Closer to 1b—the slope at point A is almost the same as the slope between points A and C since there is not much variation over a small interval. In 1a, we are looking at a large interval, which could have included many driving events.

   c. Make your best estimate of the car's instant fuel economy at Point B.

≈45 mi/gal slightly flatter than the slope between A and C

Average Rate of Change (AROC)	The <u>average rate of change of a function</u> $f(x)$ over the interval $[a, b]$ is given by: $$AROC = \frac{f(b) - f(a)}{b - a}$$ The average rate of change tells us the slope of the line through the two points $(a, f(a))$ and $(b, f(b))$. This is also known as the slope of a secant line.

Instantaneous Rate of Change	The <u>instantaneous rate of change of a function</u> at a point (a, b) is defined as the slope of the tangent line at the point (a, b).

Example 1: Let $D(t)$ represent the temperature in Denver, Colorado (measured in °Fahrenheit), t hours after midnight. 3pm

t	6	7	9	11	15
$D(t)$	68	69	77	82	71

a.) Find the average rate of change in temperature between 6 a.m. and 3 p.m.

$$ARDC = \frac{D(15)-D(6)}{15-6} = \frac{71-68}{15-6} = \frac{3}{9} = \frac{1}{3}\,°F/hr$$

b.) When did the temperature change at a faster rate, between 9 a.m. and 11 a.m., or between 11 a.m. and 3 p.m.? Justify your answer.

9am to 11am: $\frac{D(11)-D(9)}{11-9} = \frac{82-77}{11-9} = \frac{5}{2}\,°F/hr$

11am to 3pm: $\frac{D(15)-D(11)}{15-11} = \frac{71-82}{15-11} = -\frac{11}{4}\,°F/h$

Since $\left|-\frac{11}{4}\right| > \left|\frac{5}{2}\right|$, the temperature changed at a faster rate from 11am to 3pm

c.) Estimate the instantaneous rate of change in temperature at 9 a.m.

Use the times closest to 9 am:

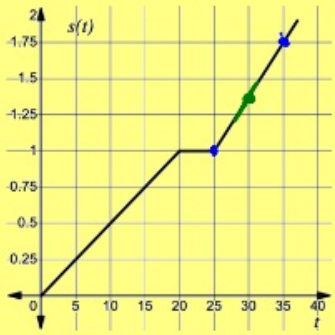
7am to 11am: $\frac{D(11)-D(7)}{11-7} = \frac{82-69}{11-7} = \frac{13}{4} = 3.25°F/hr$

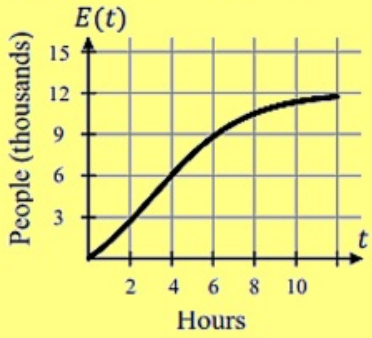
Example 2: The graph of s models Moriah's walk to school, where $s(t)$ is her distance from home, in miles, t minutes after leaving her home. Estimate Moriah's speed at $t = 30$ minutes.

Estimate the slope of the tangent line at $t = 30$ min (the slope of the line segment)

$$\frac{s(35)-s(25)}{35-25} = \frac{1.75-1}{10} = \frac{0.75}{10}$$

$$= 0.075 \text{ mi/min}$$



You Try: The number of people who have entered an amusement park is modeled by the function E, where $E(t)$ gives the number of people in thousands who have entered the park and t gives the number of hours since 10:00 a.m. for $0 \leq t \leq 11$. The graph of the function E is shown to the right.

Give a rough estimate of the instantaneous rate of change at $t = 3$.